Kafka, Data Streams & Event-Driven Architecture

How to connect SAS Viya with data streams

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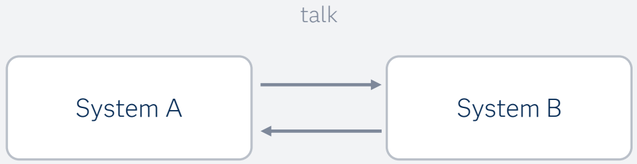
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Phone call, newspaper, radio

Three ways systems talk.

Phone call

= a request/response API

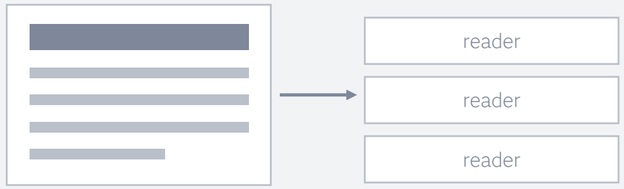


both on the line at the same moment

- Two-way: a question, then an answer (synchronous)
- One caller, one listener
- Both sides must be free at the same moment

Newspaper

= batch

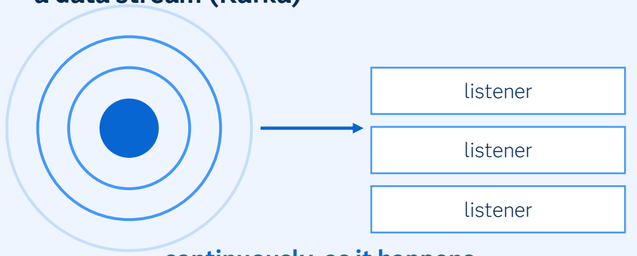


printed once, at 06:00

- One-way, to every reader at once
- Arrives on a clock, not when the news breaks
- Always yesterday — miss it and you wait a day

Radio station

= a data stream (Kafka)



continuously, as it happens

- One-way: to every listener at once (asynchronous)
- The station does not know who is listening
- Any number can tune in, each on its own

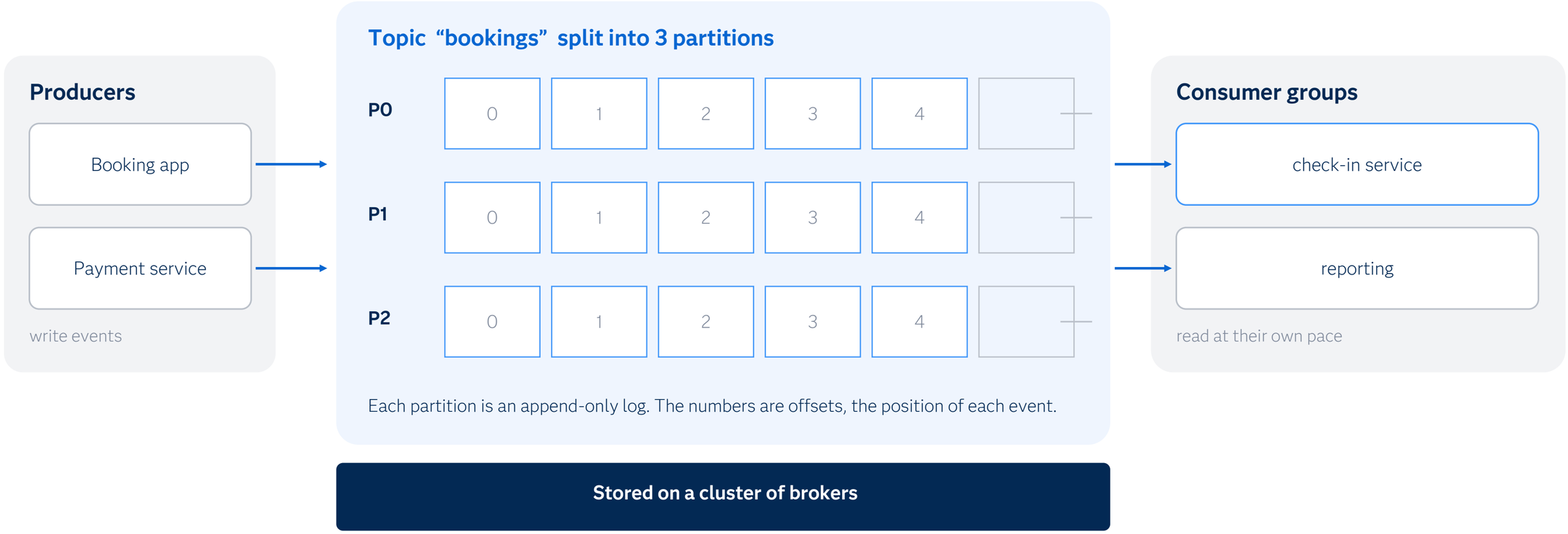
Kafka is a radio station with a recorder.

The broadcast is stored: a listener who tunes in late, or a new one next year, can still play back everything from the start.

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How Kafka is built

Producers write, brokers store, consumer groups read.



Producer

Writes an event to a topic. Does not know who will read it.

Partition

An ordered log. Events with the same key land in the same one.

Broker

A server storing partitions, with copies on other brokers.

Consumer group

Reads the topic and keeps its own position, reading deletes nothing.

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Where it is used

Five examples each from aviation, banking and insurance.

Aviation

1. Flight status and delay alerts pushed to passengers the second they change
2. Aircraft sensor telemetry streamed for predictive maintenance
3. Baggage scan events tracked end to end across airports
4. Crew and gate re-planning the moment a rotation breaks
5. Seat inventory kept in sync across every booking channel

Banking

1. Fraud scoring on a card authorisation, inside the payment window
2. Instant payments processed and confirmed in seconds
3. Balance updates and push notifications after every transaction
4. Market and trade data distributed to many desks at once
5. An immutable event log as the audit and regulatory-reporting trail

Insurance

1. Telematics data from vehicles feeding usage-based motor pricing
2. First notice of loss starting the claims workflow immediately
3. Anomaly detection on claims as they are submitted
4. Quote and underwriting decisions scored in real time
5. One policy change fanned out to CRM, billing and analytics at once

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SpeedyStore reads Kafka itself

The database subscribes to the topic. Nothing in between.

Kafka topic

events arriving all the time

→

SAS SpeedyStore

It subscribes to the topic

and stores every event as it arrives.

```
CREATE PIPELINE …  LOAD DATA KAFKA …
```

→

Dashboards, reports, models

querying the same table

There is no program in between. The database is the consumer.

1 Nothing extra to run

No separate application, no extra server to look after.

2 Nothing lost, nothing doubled

Every event lands exactly once, even if something crashes.

3 Usable straight away

The data can be queried the moment it arrives.

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A small helper in between

When Viya cannot read Kafka itself, something else does it.

Kafka topic

events arriving all the time

→

A small helper

It listens to the topic,

collects messages,

and writes them into Viya.

→

SAS Viya CAS table

in memory, ready in Studio and Visual Analytics

It only reports “done” once the data is safely in Viya.

1 A small helper, nothing more

An ordinary container, the kind your IT team already runs.

2 It writes in batches

It collects a few hundred messages and sends them together.

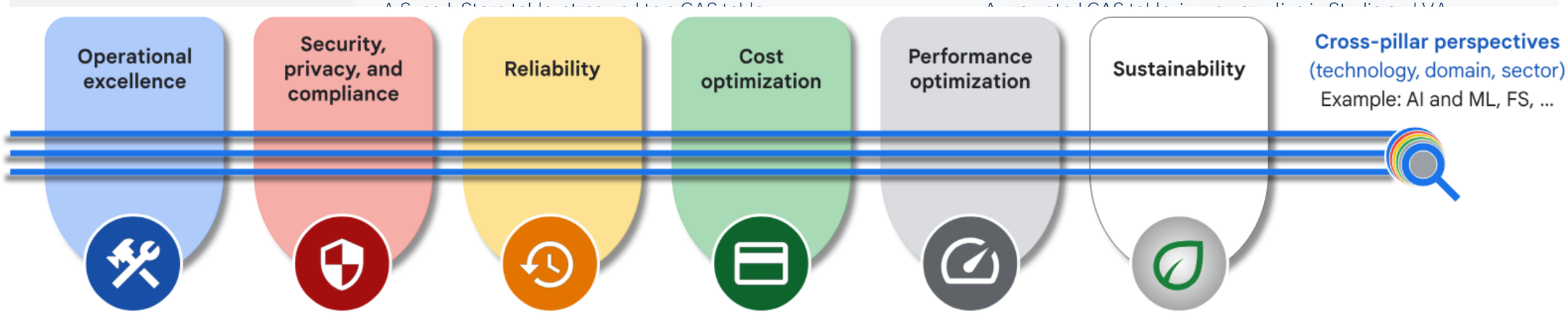
3 Nothing gets lost

If it crashes it resumes, so a batch can arrive twice.

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Which route into Viya?

Two ways to land a Kafka stream, and when each one wins.

Decision driver	SpeedyStore pipeline	swat-streaming bridge
	A SpeedyStore table streamed into a CAS table	A user-created CAS table in your application in Studio or VA
		
	Quelle: Google Cloud	
Maintenance	None beyond the database itself	An image to build, patch, monitor and be responsible for

Default to the pipeline when SpeedyStore is in the estate and the data belongs in the database.
Choose the bridge when the result has to be a CAS table, when Viya cannot reach Kafka natively, or when the work in flight is more than a mapping.

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Demo

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Thank you!

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